



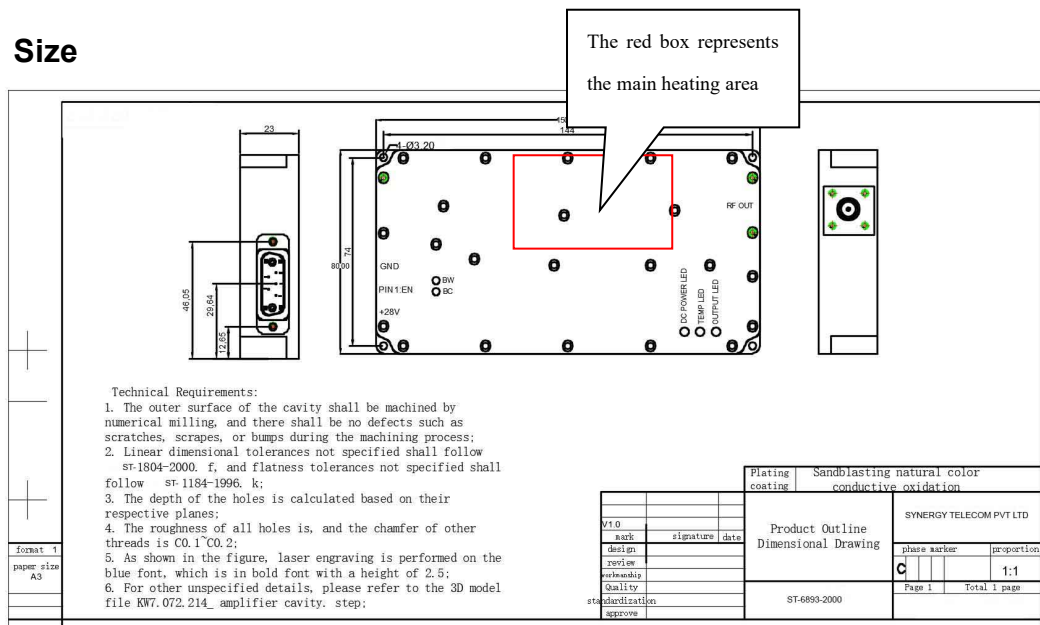
## 100W Module Technical Datasheet

### 1.RF DATA

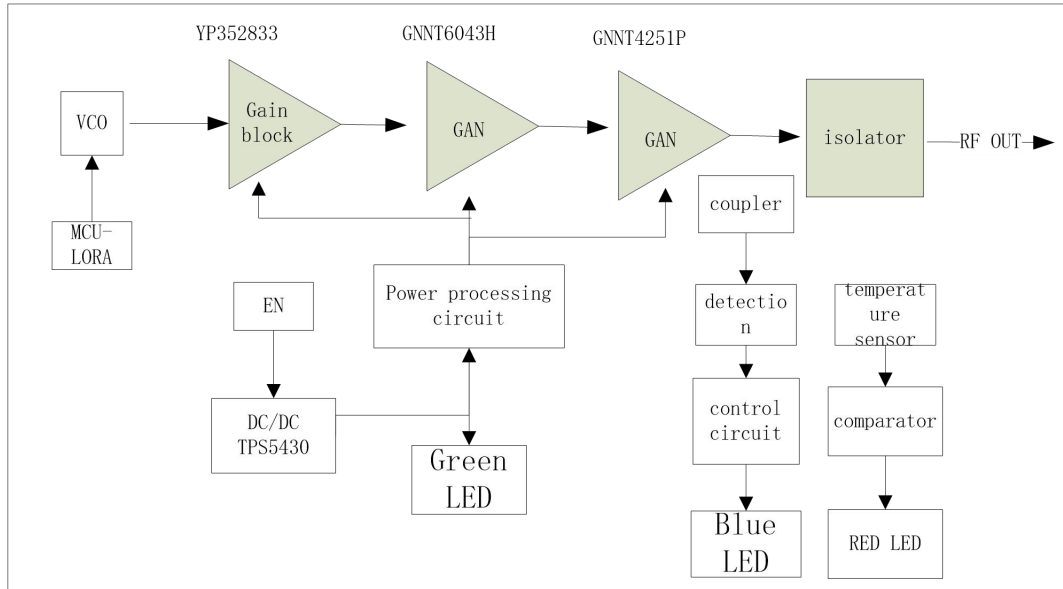
| Item                        |                     | Spec.                            | Remarks                                                                                               |
|-----------------------------|---------------------|----------------------------------|-------------------------------------------------------------------------------------------------------|
| Frequency Range             |                     | 3600-3800MHz                     | Bandwidth Range±10MHz                                                                                 |
| Working Voltage             |                     | 28V                              | 24-30V                                                                                                |
| Analog sweeping speed       |                     | 120K                             | 50<br>50KHz-300Kcustomizable                                                                          |
| Output Power (Max)          |                     | ≥50 dBm                          | 100W                                                                                                  |
| Gain (Max)                  |                     | 50±1 dB                          |                                                                                                       |
| Ripple in Band              |                     | ≤ 3 dB                           | Peak                                                                                                  |
| Bandwidth adjustment        |                     | Yes                              | By screw                                                                                              |
| Center adjustment           |                     | Yes                              | By screw                                                                                              |
| Enable control              |                     | Customize                        | 3-28V                                                                                                 |
| EN switch frequency         |                     | 10000 times                      | Switching time interval ≥ 3S                                                                          |
| temperature protection      |                     | Turn off the amplifier at 100 °C |                                                                                                       |
| Output VSWR                 |                     | ≤2                               |                                                                                                       |
| High Low Temperature test   | Working Temperature | -30~+85°C                        | Working properly                                                                                      |
|                             | Gain Stability      | ±1 dB                            | at<br>-20°C~+55°C                                                                                     |
|                             | Power stability     | ±1 dB                            | at<br>-20°C~+55°C                                                                                     |
| Working current             |                     | ≤9A @+28V dc;                    | output 100W                                                                                           |
| Power supply Port           |                     | Positive and negative lines      | 24-28V;<br>GND                                                                                        |
| Assembly execution standard |                     | IPC-A-610 Class 1                |                                                                                                       |
| Design execution standard   |                     | ST 20668-1998                    |                                                                                                       |
| RF output connector         |                     | N-Female                         |                                                                                                       |
| Green LED indicator light   |                     | CD Power ON/OFF                  | When the power supply voltage is normal, it lights up, and when the power is turned off, it turns off |
| Blue LED indicator light    |                     | RF Power ON/OFF                  | Turn off when the output power is less than 5-10dBm, and turn on when the output power is normal      |

| Item                    | Spec.                             | Remarks                                                                     |
|-------------------------|-----------------------------------|-----------------------------------------------------------------------------|
| Red LED indicator light | Temperature alarm indicator light | temperature is above 100 degrees, it lights up but does not turn off the PA |
| Working life            | 10000 hours                       | The temperature of the chamber during operation shall not exceed 65 °C      |
| Dimension               | 150*80*23 mm                      |                                                                             |
| Installation Dimension  | 144*74*23 mm                      |                                                                             |
| Screw Size              | 4-Ø3.20 mm                        |                                                                             |
| Weight                  | 590g                              |                                                                             |

## 2. Size



- 3.The red box indicates the location of the low-level amplification tube, which is the main heating area with a heat generation of approximately 180W;
- 4.The cavity uses copper blocks for heat conduction, evenly transferring heat to the aluminum cavity;
- 5.Note that when the red LED lights up, check if there are any issues with the heat dissipation, if the screws securing the PA are tightened, and if the PA and heat sink are coated with heat dissipation silicone grease
- 6.Technical Solution:



## 7、Internal product images

